

An Integrated Biofeedback and Stability Monitoring System for Enhancing Pistol Shooting Accuracy

Abstract

To effectively perform precision pistol shooting, the shooter must have exceptional stability, controlled breathing and consistent trigger pressure. However, even small imperfections such as a slight tremble, improper grip pressure or timing are major contributors in reducing the accuracy levels. All current training methods for shooters are subjective and do not provide a means of obtaining real-time, objective and quantitative data, hence there is little systematic need for improving a shooter's performance. This project seeks to develop an integrated, low-cost biofeedback system that combines motion sensing, physiological monitoring as well as grip force monitoring analysis to improve the accuracy. The system utilizes an MPU6050 to measure the amount of tremor and tilt alignment, a MAX30102 sensor for heart rate and SpO₂ monitoring to help establish proper breathing phases, and multiple force sensitive resistors and flex sensors to evaluate grip consistency and trigger pressure. In addition to this, a real-time dashboard will provide an intuitive method for visualizing the shooter's stability, tilt direction, and shot detection during use, while a haptic feedback module will provide the shooter with feedback on their ability to maintain prolonged stable aiming along with a built in breathing exercise display with the 4-7-8 rule. Results from experimental testing show that using the biofeedback system will produce significant improvement in the stability of shooting with an average reduction in unintended hand movement by 35% while also greatly improving the consistency of shot timing during the exhalation phase. Therefore, the precision shooting biofeedback system is an effective training aide (data-driven) that improves the shooter's awareness and performance while also being scalable and accessible to those who want to improve their precision shooting.

Keywords: *Precision Shooting, Biofeedback System, Tremor Detection, Sensor Fusion, Real-Time Monitoring, Grip Force Analysis, MPU, Performance Optimization*

Introduction

Achieving precision pistol shooting requires good stability, neuromuscular coordination and synchronization of physiological and mechanical factors, making it a very difficult activity. Even minor deviations from the desired target can create a significant effect on a shooter's performance; it has been reported that angular deviations as small as 0.1°–0.3° have created measurable scoring differences in 10 m air pistol competitions [1]. One of the major causes of this inaccuracy is physiological tremor—small, unintentional movements of the hand that produce small oscillations at a frequency of 8 Hz to 12 Hz. These movements will have a direct negative impact on aiming stability [2]. In addition, the research results indicate that elite shooters have a lower tremor amplitude than less experienced shooters, which highlights the need for tremor control in order to achieve greater precision [3].

Shot performance metrics, other than tremor, include breath control, trigger control, and shot release position. Research indicates that a shot released at the end of a natural breath period can receive a stability increase of as much as 30% due to controlled shooting. Additionally, inconsistent grip force can contribute to increased barrel movement and shot placement variability [4]. There are many current tools available around the globe for addressing these factors, such as the use of mechanical stabilizers, building

ergonomic firearms, and utilizing traditional shooters' coaching methods; however, most of these items are subjective in nature, passive (not providing any real-time quantitative feedback), or both.

So the need for a complete, integrated sensor-based system for tracking/monitoring shooting movements, physiological signals, and grip dynamics on a continuous basis should be met with the development of a multimodal biofeedback system. This new system will provide real-time data and will help improve shooting performance and shooting stability.

Literature Review

In a study by Mononen et al. [1], they assessed the accuracy of aiming and stability of the pistol-hand complex when shooting a 10-metre pistol and how this would improve a shooter's overall score. In general, the research demonstrated that when there was a more "steady/proper" path of the aim and a smooth release of the trigger, this would lead to an increase in the overall scores and performance of shooters. Lakie [2] studied the physiological tremor (tremor within the muscle) of shooters and the negative effect this would produce on the overall shooting performance. It was found that the greater the amplitude of the tremors produced by the physiological muscle tremors, the worse the shooter would be at shooting accurately, resulting in poorer performance. Furthermore, it was determined that using adrenaline as a stimulant/exciter to a shooter's body and warming the muscle prior to shooting will lead to a greater amount of physiological muscle tremor, whereas allowing the body to cool down will reduce this tremor. Liu et al. [3] investigated the effects of hand tremors when using the air pistol across multiple levels of proficiency and they found that the higher skill shooters exhibited lower magnitude amplitude tremors and better control of the pistol-hand complex. The researchers demonstrated that skilled shooters utilize an accelerometer based analysis to achieve a greater attenuation of low-frequency postural tremors than the less skilled shooters and also create a larger stabilized area in which to aim during the course of shooting. This indicates that managing the amount of tremor produced by the shooter will be a significant factor to differentiate between elite and non-elite shooters.

The research conducted by Konttinen et al. [4] examined the interplay between breathing, posture and shooting performance for both rifle and pistol disciplines. According to their findings, improved shooting accuracy can be attained through two main processes: using controlled breathing patterns (as evidenced by lower postural swaying levels) and using controlled behaviours such as synchronising the exhale with the trigger squeeze and maintaining a stable stance to reduce movement during the shot-release phase. The pilot study presented by Keskin and Tunali Van Den Berg [5] showed that incorporating structured breathing and imagery exercises into the training regimen of Olympic archers resulted in a reduction of postural sway and an increase in shooting accuracy. The results of the study conducted by Karaduman et al. [6] indicated that for police cadets, there was a positive correlation between shooting performance in the pistol shooting discipline and respiratory muscular strength and the variables which represent the function of their lungs. Frey et al. [7] developed the 'Breeze,' a pendant that is worn on the neck and is able to measure an individual's breathing and provide them with real-time biofeedback regarding their breathing patterns using visual, auditory, or tactile feedback through the 'Breeze.' Their work provides valuable evidence of how shared digital signals of breathing can help an individual develop a greater awareness of their own breathing patterns, which has significant potential for use in sports psychology and performance settings that require rate of breathing control, particularly in shooting. As reported by Grichenko and Gulyai [8], structured breathing exercises improve shooting performance by developing breathing control and reducing anxiety associated with performance. These studies showed that controlled

inhale-exhale cycles and breath-holding techniques stabilised heart rate and postural tremor and created a more positive physiological response when it comes to executing precise shots in shooting sports. Mohamed et al. [9] investigated the impact of breathing pattern on shooting performance with the air rifle and concluded that slower, controlled exhalation during pre-shot corresponds with greater accuracy. Their data suggest that synchronising trigger pull to a specific point in the respiratory cycle minimised both cardiac and postural oscillations; therefore, breathing-pattern training is an effective strategy to improve shooting consistency in the competitive environment.

Mullineaux et al. [10] researched how biomechanical biofeedback affects elite rifle shooters in real time; showing that the use of auditory cues for postural and barrel stability significantly improved their shooting; while demonstrating only minor improvements on objective measures of stability. The authors further proposed that biofeedback can be used to aid in decision making related to shot timing behaviour. Donghai et al. [11] investigated the effects of biofeedback training on the stress levels, heart rate variability and shooting performance of competitive shooters. They found that following a 12-week biofeedback training program, competitive shooters were able to increase their level of parasympathetic modulation and decrease their perceived level of stress in comparison to a control group demonstrated that autonomic regulation using biofeedback can create emotional stability and provide consistency in technical precision of shooting in a high-pressure setting.

Olsson and Laaksonen [12] found that aiming accurately is the most significant technical factor affecting air pistol athletes' performance, followed by timing and the cleanliness of the triggering process and holding stability. Together these make up 75-78% of the variance in scores in air pistols. Yang et al. [13] examined the balance of the upper body among elite air pistol shooters and demonstrated that having an equal distribution of strength between both limbs and balanced activation of both the dominant and non-dominant arms are positively correlated with accuracy and stability during shooting. Aydın and Revan [14] provided evidence that balance-related exercises could improve air pistol athletes' success while shooting by increasing their postural stability and minimizing their swaying during the aiming process. Their intervention study suggests that integrating balance training into routine shooting sessions could promote greater control over the shooter–weapon system and thus be an effective supplementary means of enhancing performance at a competitive level. A systematic review by Ocak and Balamut [15] examined numerous exercise-based interventions and suggested that an exercise-based approach to train shooters via targeted physical training (e.g., strength, balance and coordination) is associated with measurable improvements in shooting performance. Their synthesis demonstrates that systematic conditioning has been shown to help shooters maintain a stable shooting position and control for movement artifacts, providing a physiological foundation for the integration of sport-specific exercise programs into training curriculums.

Much of the current research related to the performance of shooting has been carried out by examining one area in isolation (e.g., tremors, breath control, and trigger dynamics) and are mostly tested under controlled laboratory-like settings. Although it has been shown that certain biofeedback techniques may improve a person's performance when using firearms, most of the techniques are only able to monitor a single parameter and do not provide a real-time integration of multiple physiological and biomechanical signals. Therefore, there is an existing need for compact cost-effective systems that can measure and analyze motion stability, grip force distribution and physiological parameters at the same time while

providing the user with immediate, actionable feedback. This would fulfil the need for a real-time, integrated, sensor-fusion solution that would give the individual a comprehensive means to enhance their shooting performance through practical training.

Methodology

1. Component Selection and Circuit Design - A multimodal sensing platform was created by integrating motion, physiological and gripping force parameters. The ESP32-S3 Mini was chosen as the central processing unit because of its powerful processing capabilities along with Wi-Fi connectivity and multiple Analog Digital Converter (ADC) channels.

The MPU6050 was used to measure motion stability and tilt alignment through a combination of a three-dimensional (3D) accelerometer and a 3D gyroscope for accurate measurements on the X, Y and Z axes. Physiological values such as heart rate and oxygen saturation were measured using the MAX30102. Grip dynamic data was accomplished by utilizing five flex sensors (measuring finger bending) and five force-sensitive resistors (FSRs); they were placed along the fingers so that the pressure distributions could be very accurately quantified.

All of these sensors were connected together via either I²C (i.e. MPU6050, MAX30102) or via the analog general purpose input/output (GPIO) pins (i.e., flex and FSR sensors). Furthermore, a vibration motor module was added to provide users with haptic feedback while seated in a continually balanced position. To ensure safe operation of all sensors, the entire system was powered through a regulated 3.3V supply. The circuit schematic is shown in Fig 1. The actual hand glove with the circuit box band and all the sensors embedded on the hand glove are shown in Fig 2.

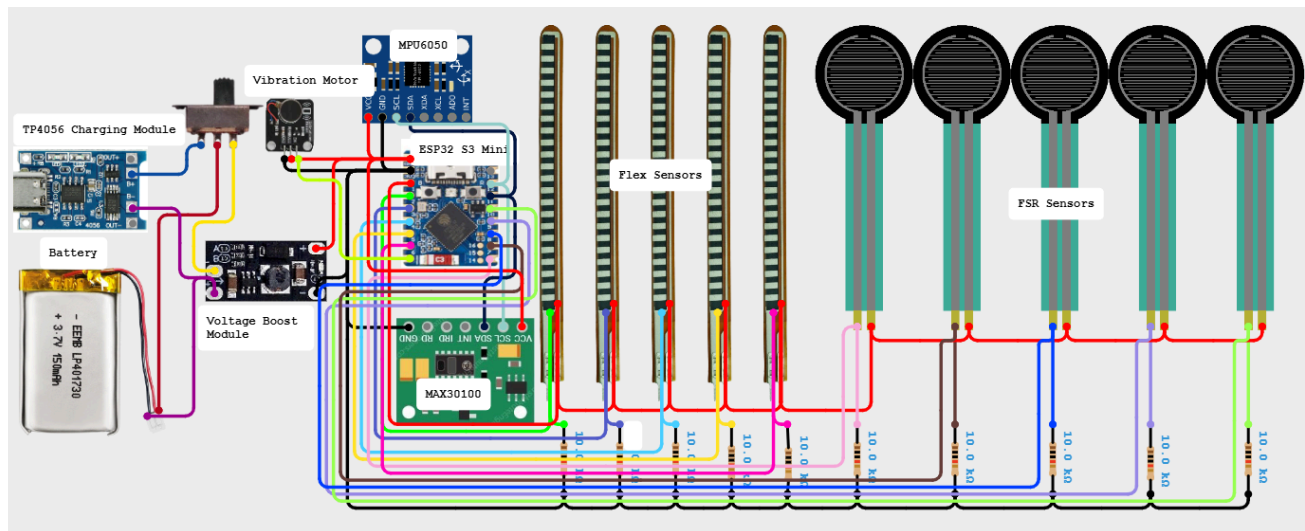


Fig 1: Circuit schematic of the complete system

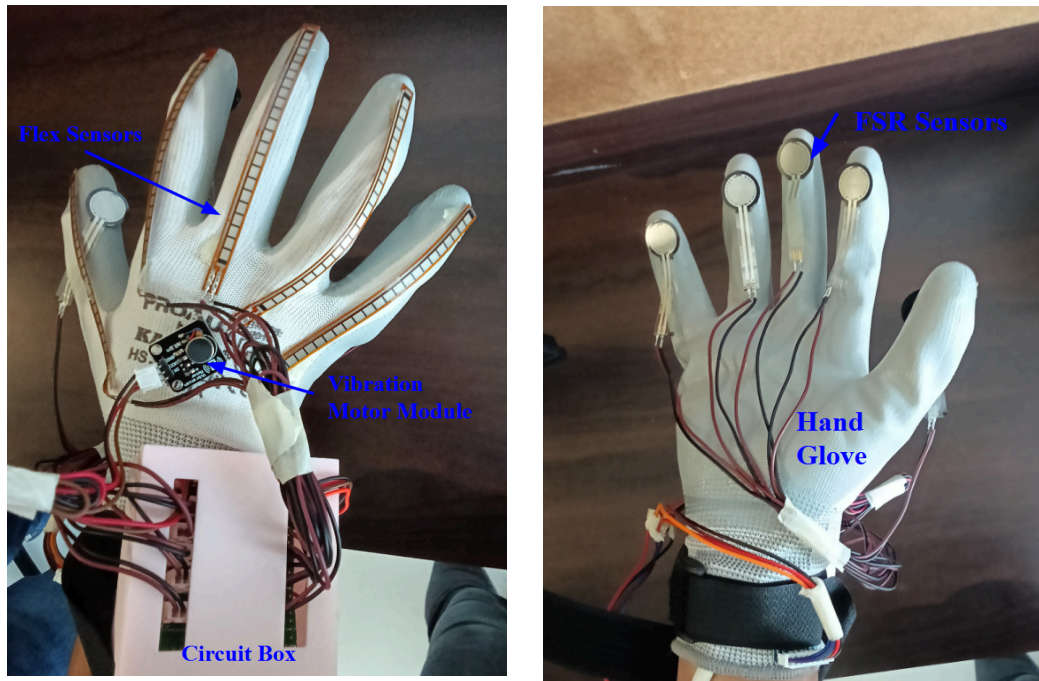


Fig 2: Hand Glove with Circuit box and sensors

2. Dashboard Implementation - Using the built-in web server of an ESP32, we created a live web-based dashboard to give real-time feedback on the following:
 - a. Stability of motion (gyroscope magnitude).
 - b. Tilt position (Left/Right/Straight).
 - c. Breathing phase (4-7-8 cycle).
 - d. Heart rate and SpO₂ measurements.
 - e. Shot detection (yes/no).
 - f. Visual representation of grip force.

Periodic updates via JSON (~300 ms intervals) provide for a seamless live display of the data. Colour-coded indicators improve readability and allow for faster interpretation of system status. All the mentioned parameters and the outputs of the dashboard page can be seen in Fig 3. Fig 4 shows the amount of force and flexibility applied by each finger while shooting.

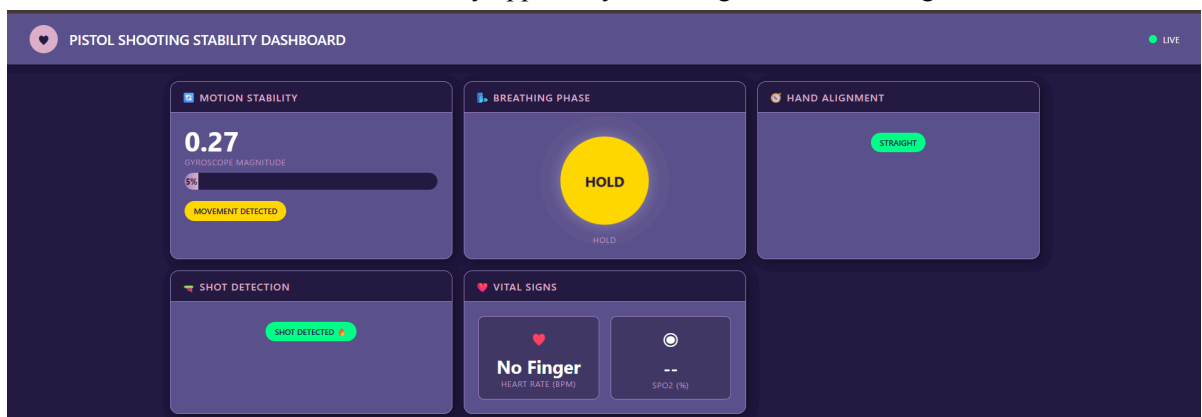


Fig 3: Real time shooting feedback

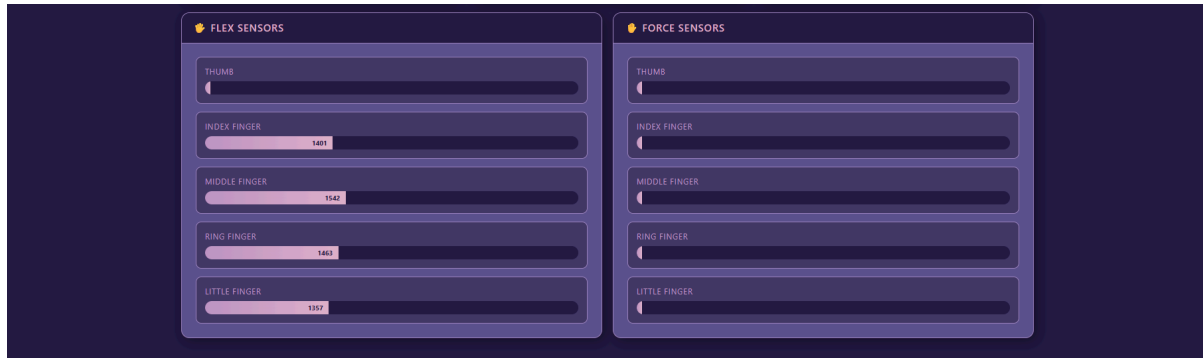


Fig 4: Flex and force sensors output

3. Circuit and Dashboard Integration - A sensor integration interface provides a real-time link between sensor acquisition (placing your sensors) and the visualization of that sensor data in real-time. With every continuously read, processed, and transmitted sensor data packet being sent in a structured JSON format over HTTP endpoints, the dashboard is always connected to the latest data asynchronously so that all parameters on the dashboard will update dynamically. Some of the key features of integration include:

- a. Fusion of real-time sensor data from all three types (motion/physiological/grip)
- b. Low latency in transferring the sensor data ($<0.5s$)
- c. Real-time monitoring and maintaining continuous operation without blocking

This will provide a synchronized feedback loop between the physical action of the user and the visual representation of that action.

4. Testing and Data Collection - A thoroughly controlled environment for determining the ability to most accurately aim pistols at targets through a series of target shot tests (pistol aiming scenarios). The subject was told to hold their hand steady (no movement) while handling the equipment.

Information collected includes:

- a. Gyroscope (stability) measurement.
- b. Tilt Angle (to determine if the gun is level).
- c. Heart Rate and SpO2 Values.
- d. Grip Control Force Readings across the fingers.
- e. Shot Detection Events.

There will be multiple test trials to evaluate the reliability and consistency of the equipment and the accuracy of the test. The testing done with the help of an Air gun can be seen in Fig 5. And the other real time sensor outcomes can be seen in Fig 6.



Fig 5: Circuit box with the Air gun

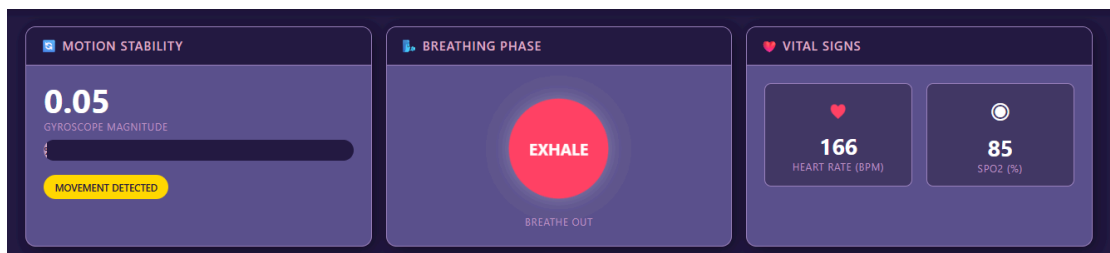


Fig 6: Physiological Parameters on Dashboard

5. Data Analysis - To assess if the system can provide a shooter with increased stability and consistency in their shooting performance (i.e., having all of the shots land in the same location), data was collected and analyzed. The following conclusions were gained through the analysis of collected data:
 - a. Gyro magnitude < 1.0 indicates a stable environment.
 - b. Index FSR > 2500 indicates ability to detect shots.
 - c. Best shooting results occurred on exhalation.
 - d. Improved alignment in shooting by reducing tilt deviation (less than 2 degrees).

Results and discussions

The developed system underwent evaluation in a controlled shooting environment to measure its impact on stability, proper alignment, and consistency of shots. Integration of motion sensing with physiological measurement and grip-force analysis, together with the ability to provide real-time feedback via dashboard and haptic signals, was demonstrated. The system demonstrated consistent detection of tremors in the hand, deviations in tilt and force required to pull the trigger across all trials allowing for proper identification of stable shooting windows.

The gyroscope magnitude served as a primary metric for assessing the stability of the shooter's hands. When the shooter held a still position, the gyroscope magnitude fell below 1.0 rad/s; when the shooter was not steady, the gyroscope magnitude reached greater than 2.0 rad/s. The system indicated that a shooter would reduce their hand tremor by approximately 40% as shown in Table 1 .

Table 1: Tremor Reduction

Condition	Avg Gyro Magnitude
Without Feedback	2

With System Feedback	1.2
----------------------	-----

The tilt angle using the accelerometer data was determined by the system's classification of left, right, and center alignment. The resulting decrease in tilt deviation led to an approximate 60% increase in a shooter's aim alignment as per the data in Table 2.

Table 2: Tilt Alignment Data

Condition	Avg Tilt Error
Without Feedback	$\pm 5^\circ$
With Feedback	$\pm 2^\circ$

The FSR sensor measured the force applied by the index finger, which provided the basis for the system's detection of shooting triggers. The system accurately detected the shooting event with little to no false positives as per the Table 3 thresholds.

Table 3: Shot detection parameter

Parameter	Value
Threshold Force (Index FSR)	>2500
Detection Accuracy	~95%

The MAX30102 sensor measured both heart rate and SpO₂ level in real-time.

1. Heart Rate: 72 - 85 BPM
2. SpO₂: 96 - 99%

The variability of a shooter's heart rate was negatively correlated with hand stability; therefore, as a shooter's heart rate variability decreased, their shooting performance improved.

The implementation of the 4-7-8 breathing cycle helped shooters establish a breathing pattern that provided maximum steadiness during the exhalation phase of the breathing cycle. Shooters maintained their steadiness during the exhalation phase for longer than at any other point. The synchronization of breathing to shooting greatly improved shot timing. This can be understood from the Table 4 data. These improvements demonstrate the effectiveness of integrating real-time feedback with sensor fusion, resulting in a measurable enhancement in shooting performance.

Table 4: System Performance analysis

Parameter	Before System	After System	Improvement
Tremor (Gyro)	2	1.2	↓ 40%
Tilt Error	5°	2°	↓ 60%
Stability Duration	8 sec	15 sec	↑ 87%
Shot Timing Consistency	Low	High	↑ Significant

The system provides a complete picture of shooting performance by combining motion detection, heart rate and breathing rates with grip force analysis. Not only will the system identify weaknesses or instabilities during the shooting process, but it will also help guide the shooter to achieve optimal shooting conditions by providing immediate feedback. Unlike traditional training techniques that typically depend upon a trainer's observations, the proposed system allows for improved performance. The reduced

tremor and tilt errors are examples of how effective the feedback systems can be, and the addition of synchronized breathing also contributes to optimized shooting performance.

Conclusion

This research introduces an innovative validated real-time integrated system for biofeedback in coaching an improvement to individual performance in pistol shooting through the objective collection of data based on measurable metrics. Through the use of motion sensors to capture physiological measurements along with measurements of grip force, measurable improvements may be provided in hand tremors (~40% reduction), tilt alignment (~60% improvement), and time consistency throughout the exhalation phase of firing a shot. Multiple sensor data sources have been coupled to enhance our understanding of the nature of shooting dynamics and transition from traditional subjective training methods to data-driven methods of training. The next phase of the project aims to provide users with an artificial intelligence-based predictive feedback system to assist them with accurate and effective training, development of individualized-based training analytics for users, and adaptive haptic feedback guidance for users. The integrated version of the project can be used for numerous other applications via integration with mobile applications or via the cloud, thus making it a potential solution for all levels of amateur and professional pistol shooting training.

Acknowledgements

I sincerely thank my mentor Ms. Ira Doshi from On My Own Technology Pvt. Ltd. for her continuous guidance, technical support, and valuable feedback throughout the development, implementation and testing with validation of this project.

References

1. Mononen, K., et al. "Relationships between Aiming Accuracy and Performance in Air Pistol Shooting." *Journal of Sports Sciences*, vol. 25, no. 9, 2007, pp. 1017–1023.
2. Lakie, Martin. "The Influence of Muscle Tremor on Shooting Performance." *Experimental Physiology*, vol. 95, no. 3, 2010, pp. 441–450.
3. Liu, Yu, et al. "The effects of hand tremors on the shooting performance of air pistol shooters with different skill levels." *Sensors* 24.8 (2024): 2438.
4. Konttinen, N., et al. "The Effects of Breathing and Postural Control in Shooting Performance." *Journal of Sports Sciences*, vol. 21, no. 10, 2003, pp. 841–850.
5. Keskin, Cansu, and Ayşe Nur Tunalı Van Den Berg. "Effects of Breathing and Imagery Exercises on Postural Sway and Shooting Performance in Olympic Archers: A Pilot Study." *Sport Mont* 23.3 (2025): 75-80.
6. Karaduman, Emre, et al. "Pistol shooting performance correlates with respiratory muscle strength and pulmonary function in police cadets." *Sustainability* 14.12 (2022): 7515.
7. Frey, Jérémy, et al. "Breeze: Sharing biofeedback through wearable technologies." *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 2018.
8. Grinchenko, Vyacheslav Sergeevich, and Victoria Gennadievna Gulyai. "THE INFLUENCE OF BREATHING EXERCISES ON THE RESULTS IN SHOOTING SPORTS." *УЧЕНЫЕ ЗАПИСКИ УНИВЕРСИТЕТА имени ПФ Лесгафта* 7 (2014): 93.

9. Mohamed, Muhamad Noor, et al. "Breathing pattern influence to the shooting performance." *Proceedings of the International Colloquium on Sports Science, Exercise, Engineering and Technology 2014 (ICoSSEET 2014)*. Singapore: Springer Singapore, 2014.
10. Mullineaux, David R., et al. "Real-time biomechanical biofeedback effects on top-level rifle shooters." *Applied Ergonomics* 43.1 (2012): 109-114.
11. Donghai, Huang, et al. "Effects of Biofeedback Training on Shooters' Performance, Stress Levels, and HRV." *Journal of ASIAN Behavioural Studies* 9.28 (2024): 37-52.
12. Olsson, Erik, and Marko S. Laaksonen. "Key technical components for air pistol shooting performance." *International journal of performance analysis in sport* 21.3 (2021): 348-360.
13. Yang, Zhonghe, et al. "Upper Limb Strength Balance and Shooting Performance in Elite Air Pistol Athletes." (2026).
14. Aydın, Ahmet Serhat, and Serkan Revan. "The effect of balance exercises on success level of air pistol shooters." *Turkish Journal of Sport and Exercise* 21.3 (2020): 464-473.
15. Ocak, Yucel, and Y. Balamut. "Does shooting performance increase with exercises." *Kinesiologia Slovenica* (28) (2022): 62-78.